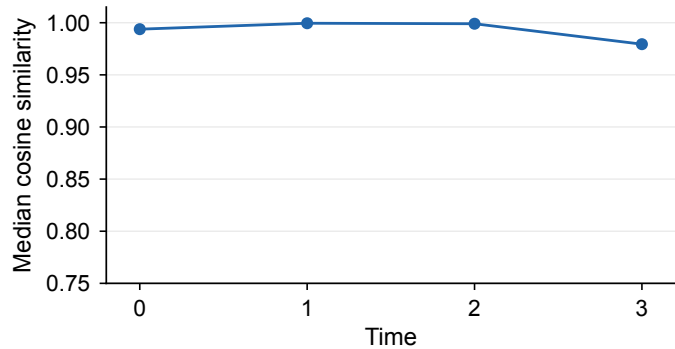
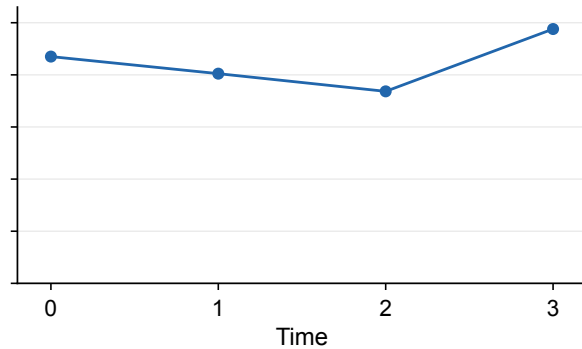


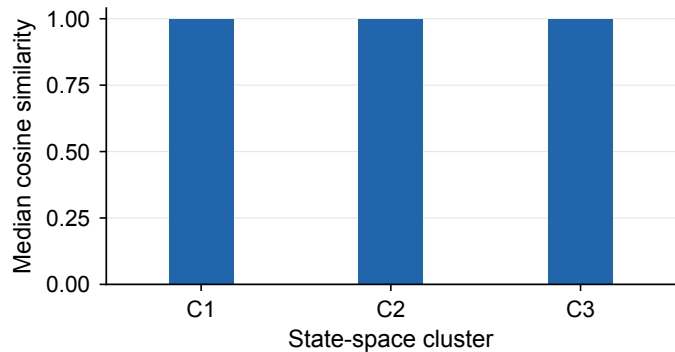
a Spatial velocity



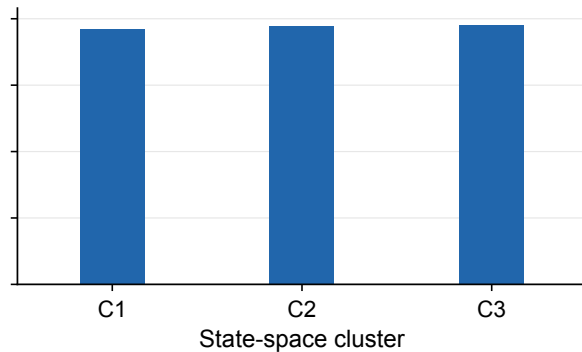
b Gene-state velocity



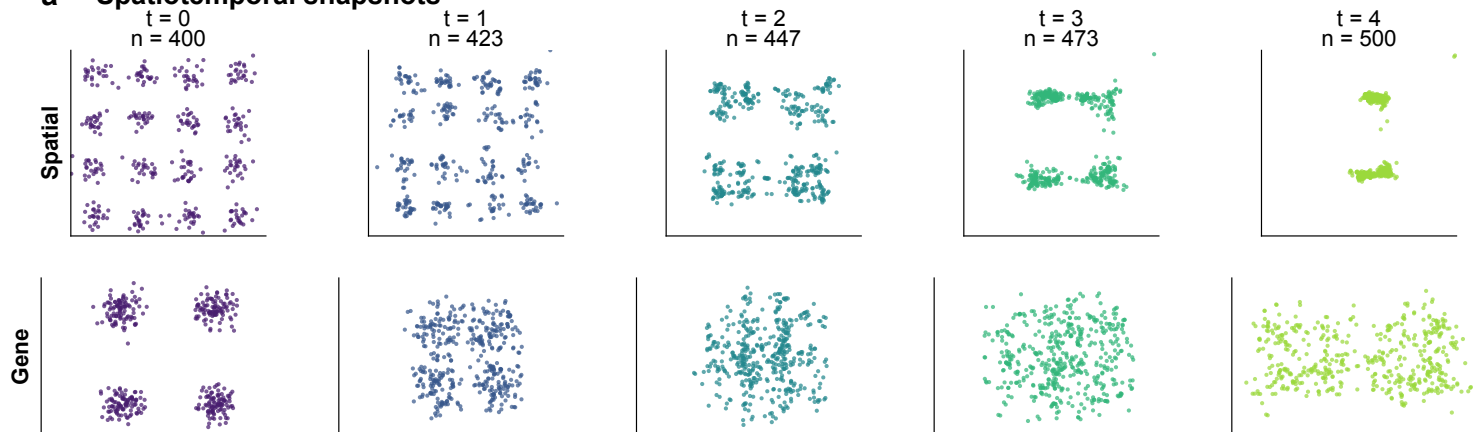
c Spatial velocity



d Gene-state velocity

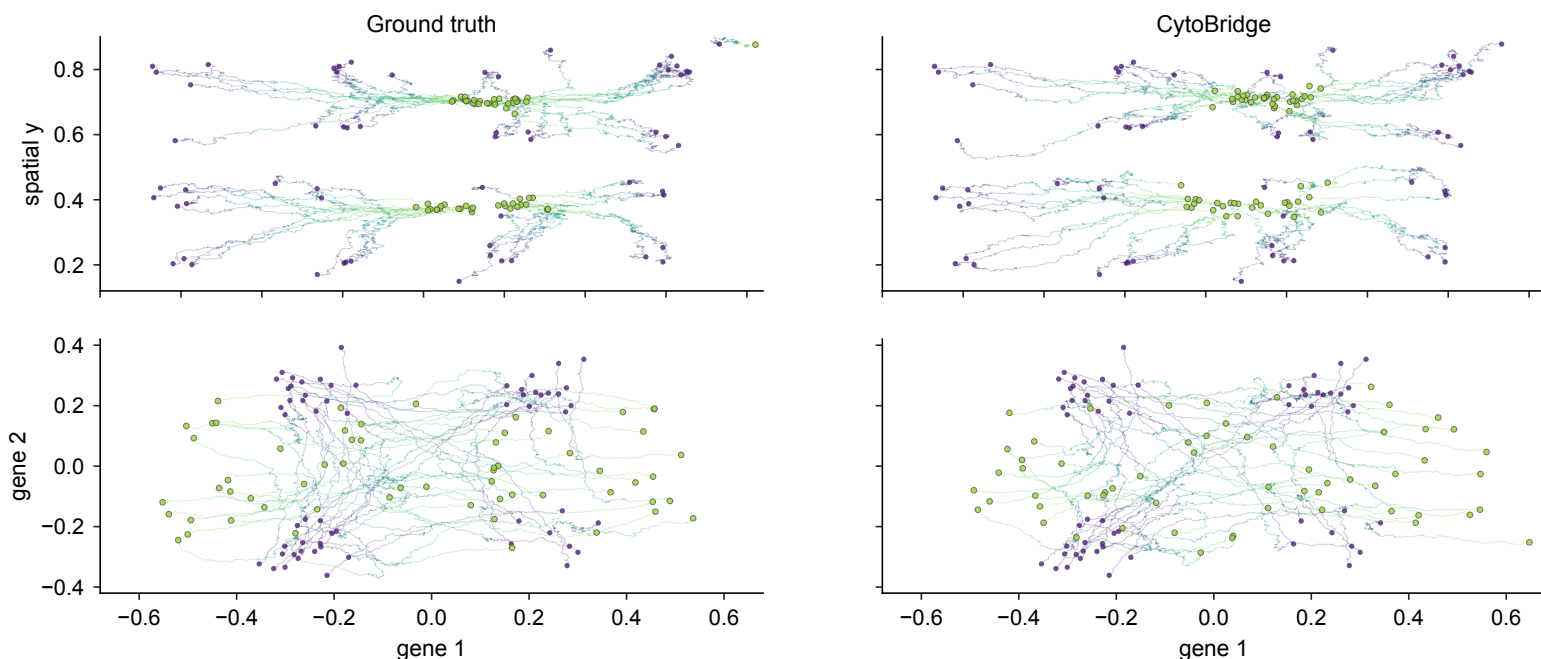


a Spatiotemporal snapshots

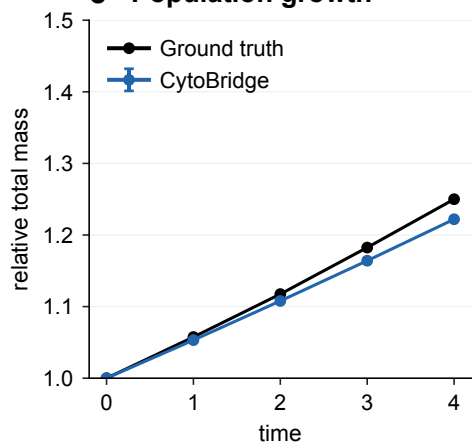


b Cell-state trajectories

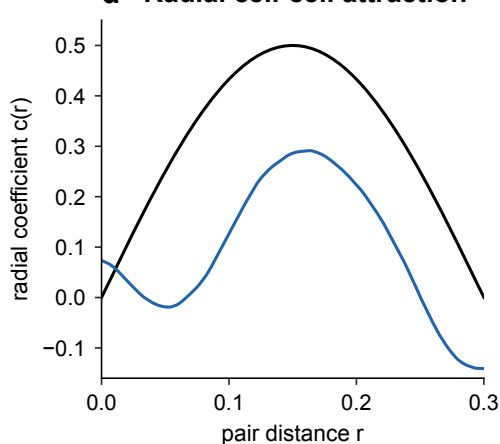
$t = 0$ $t = 4$



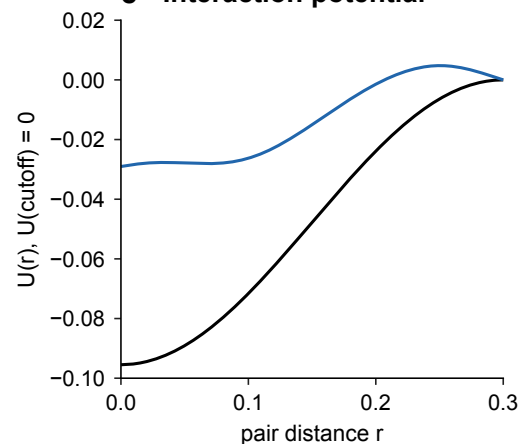
c Population growth



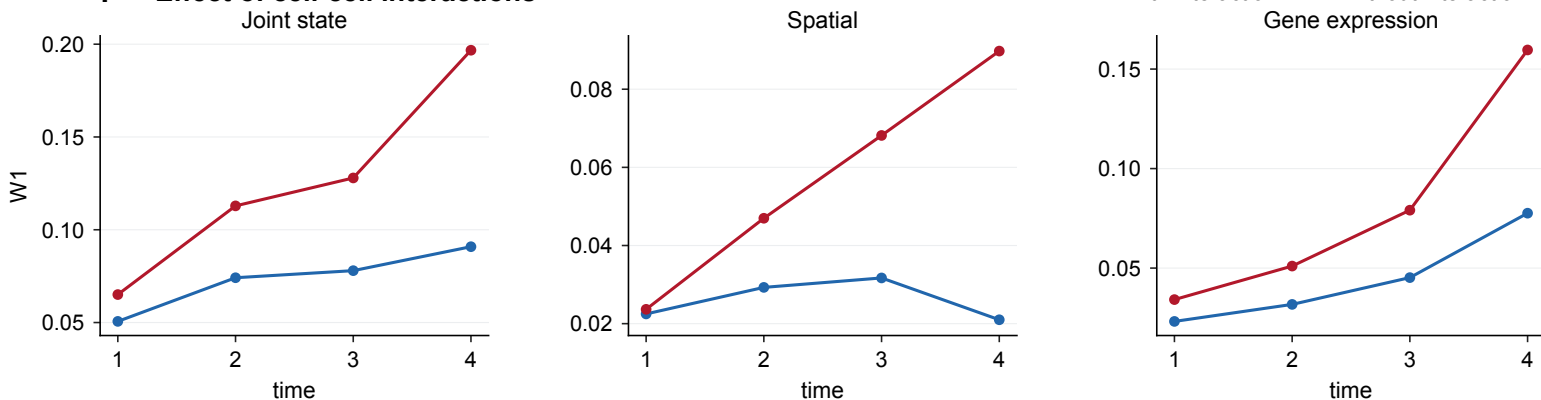
d Radial cell-cell attraction



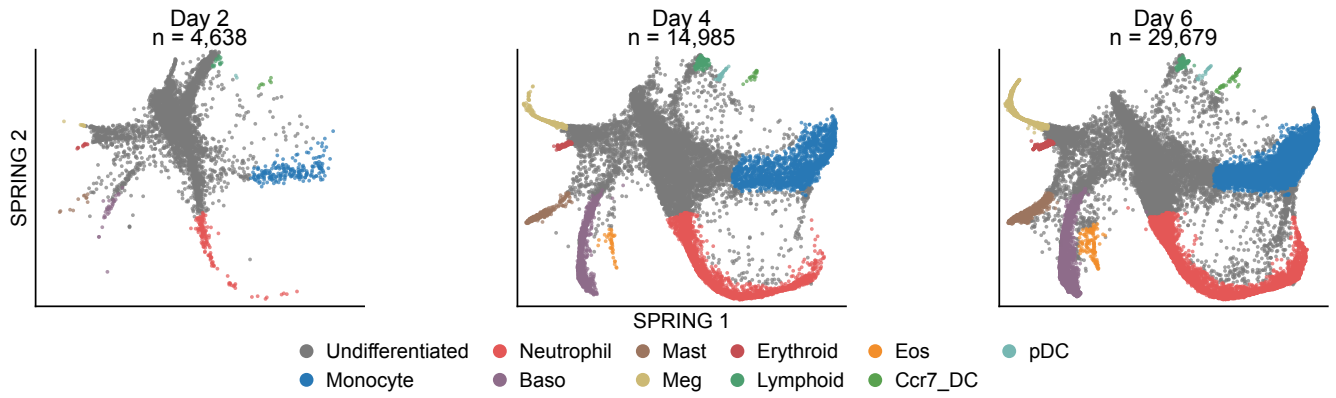
e Interaction potential



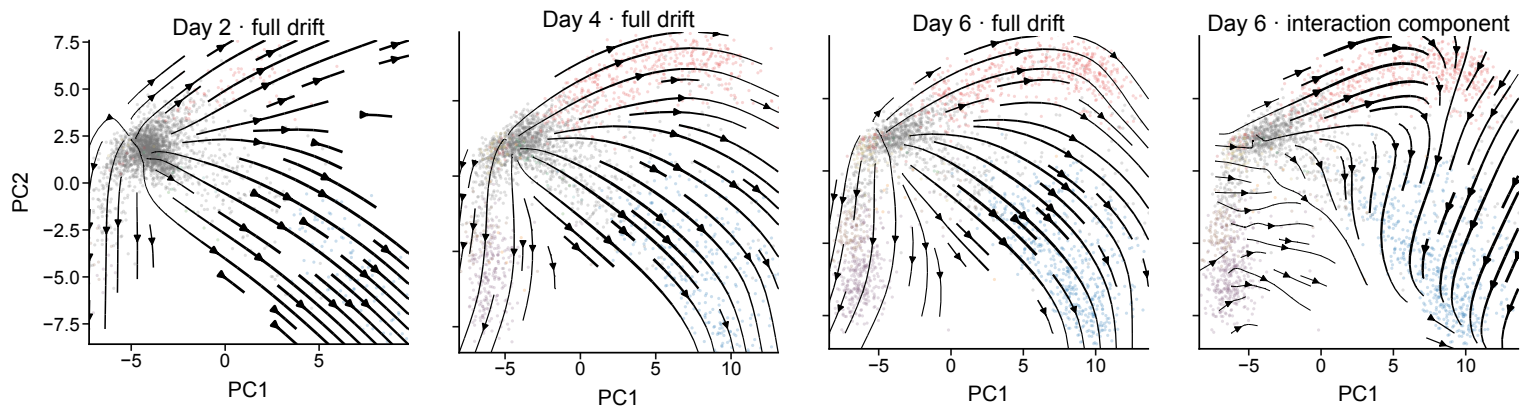
f Effect of cell-cell interactions



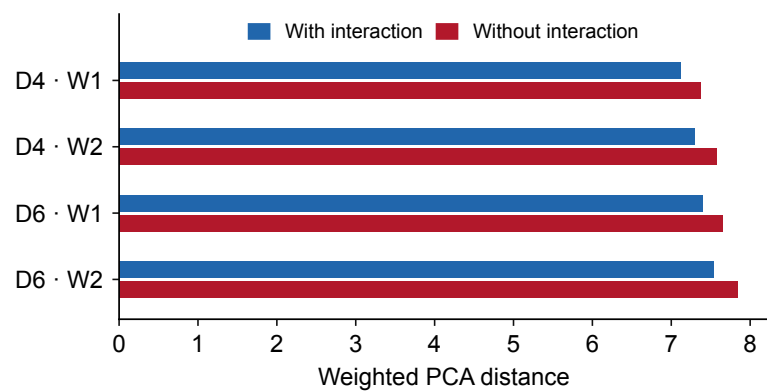
a Observed cell states across time



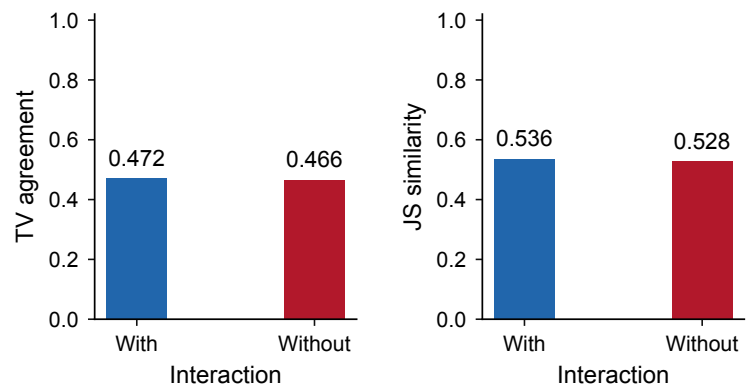
b Interaction-informed dynamics in PCA state space



c Distribution error

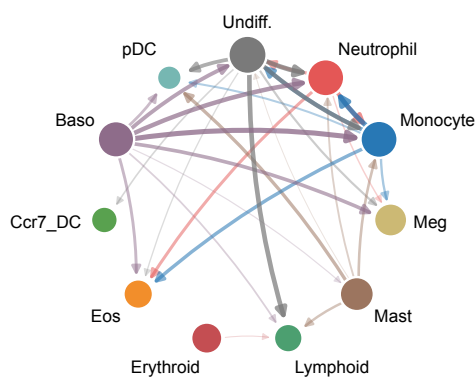


d Clone-fate agreement

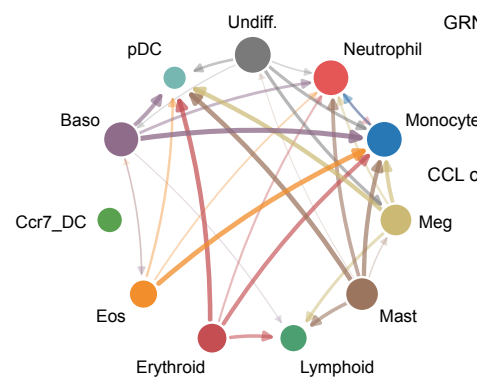


e Day 6 directed cell-cell interactions

CytoBridge interaction network

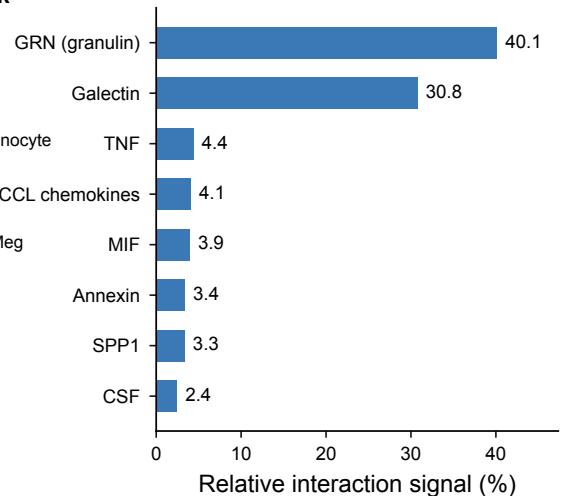


CellChat communication network

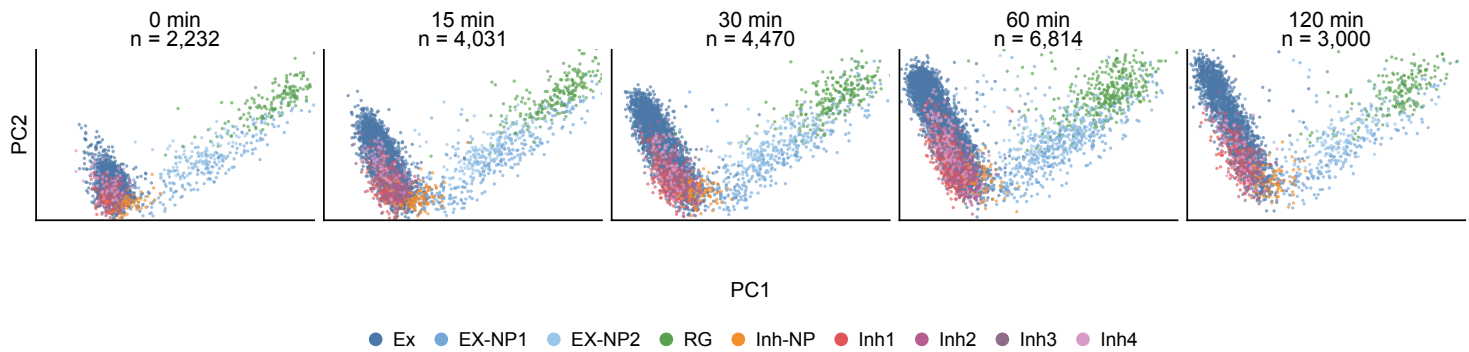


Spearman $\rho = 0.664$

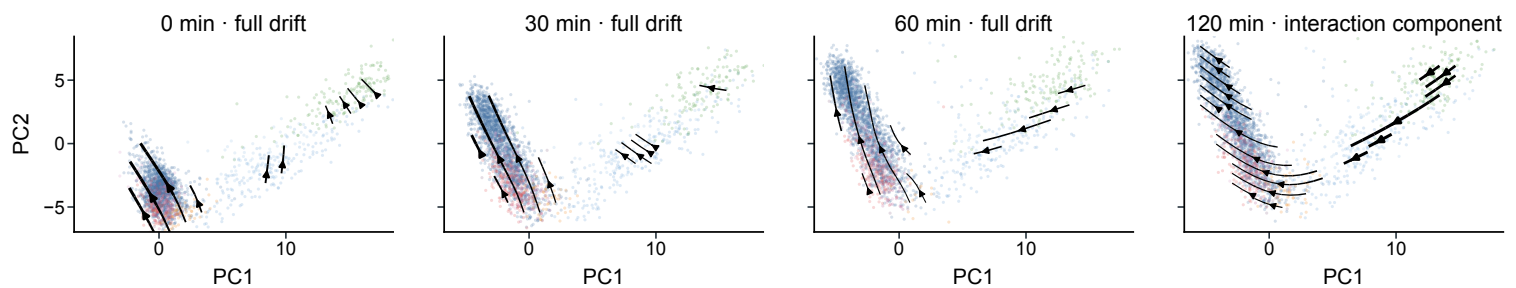
f Day 6 signaling pathways



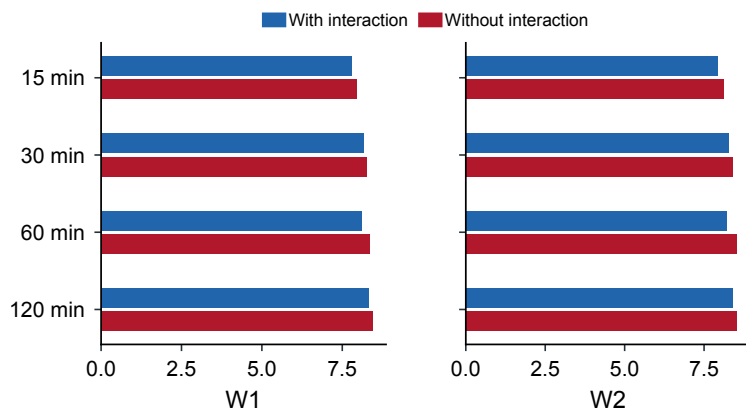
a Observed scNT cortical cell states across KCl stimulation



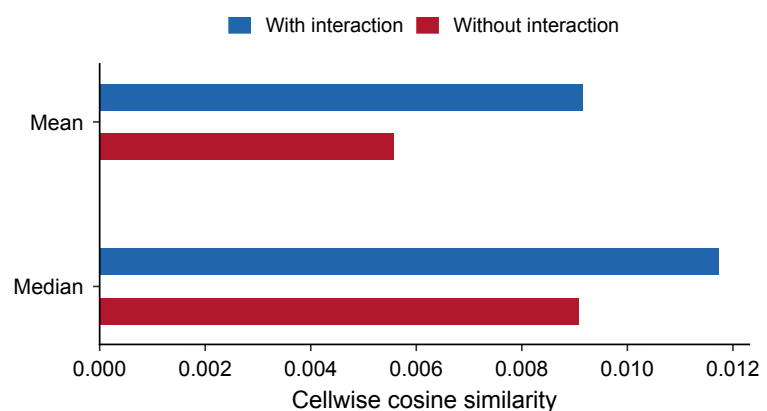
b Interaction-informed dynamics in PCA state space



c Distribution error

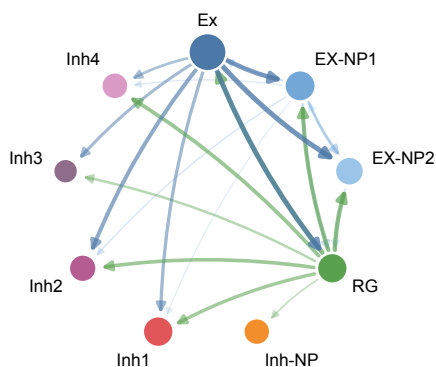


d New-RNA direction



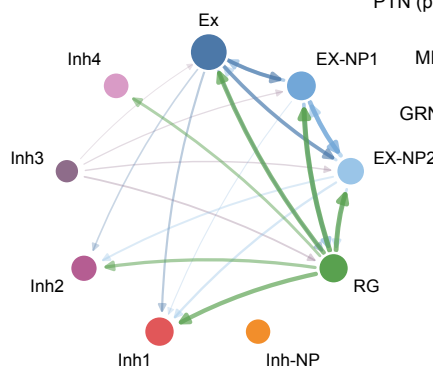
e 120 min directed interactions

CytoBridge learned message D

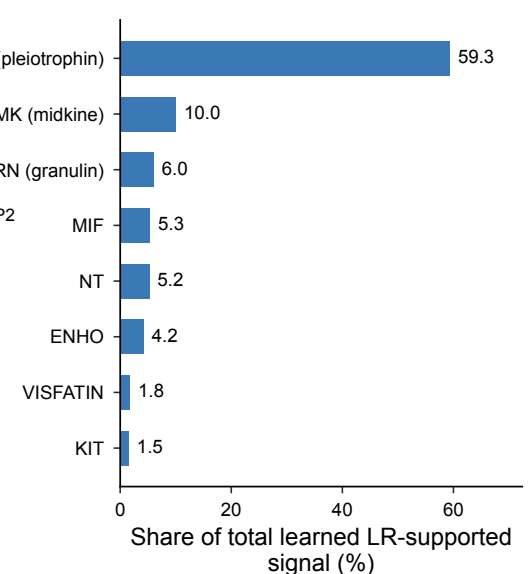


all 81 pairs · Spearman $\rho = 0.752$

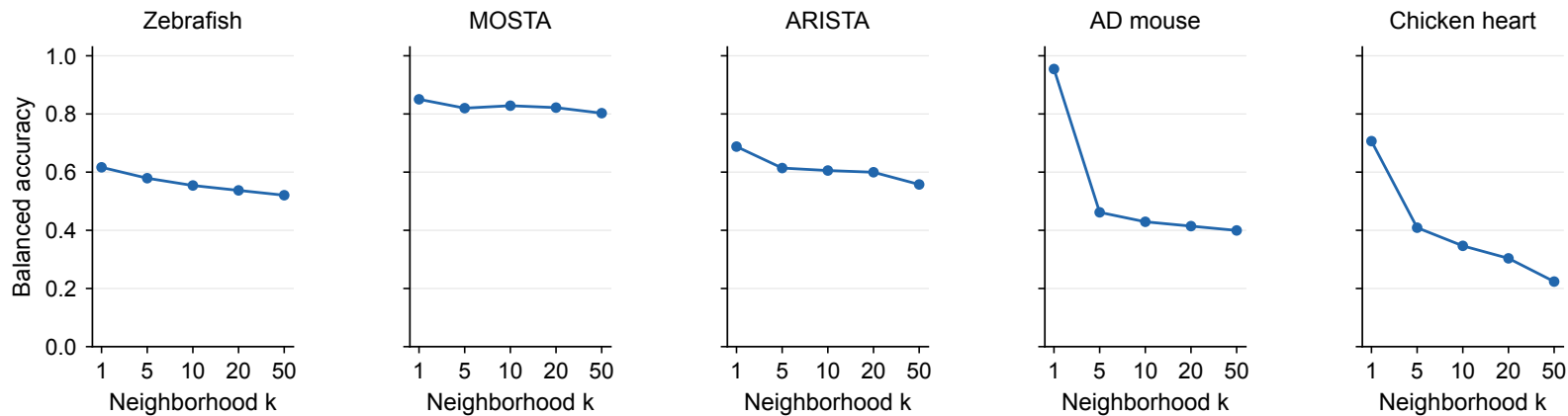
CellChat communication



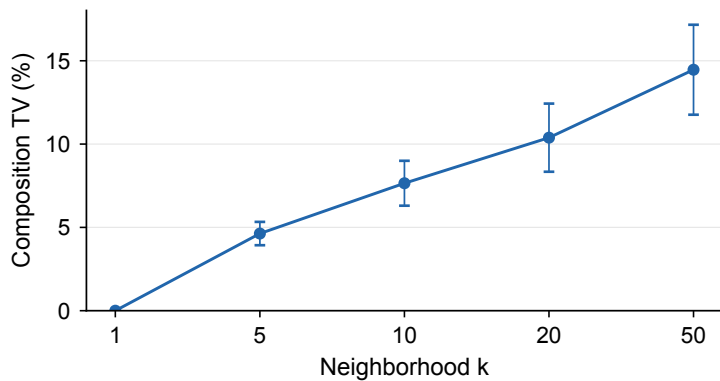
f LR-supported pathways



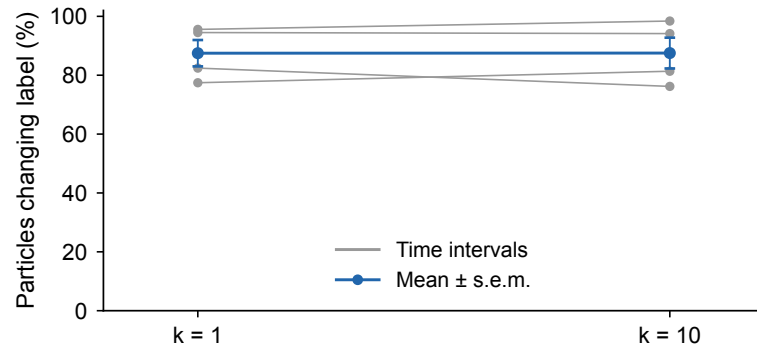
a Held-out classification



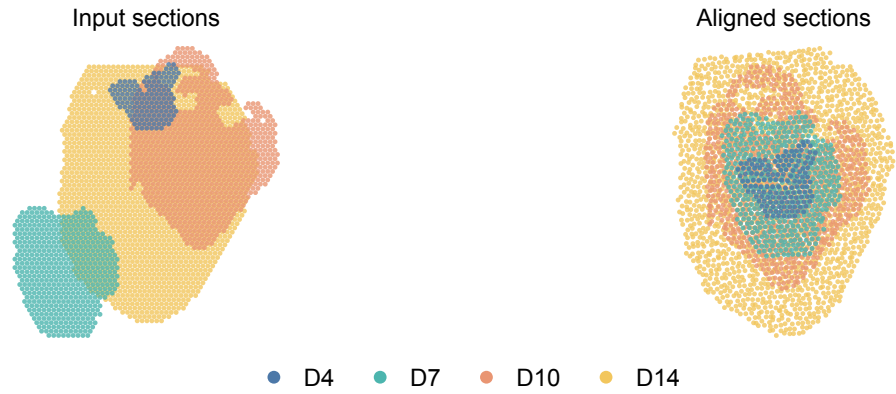
b Cell-type composition



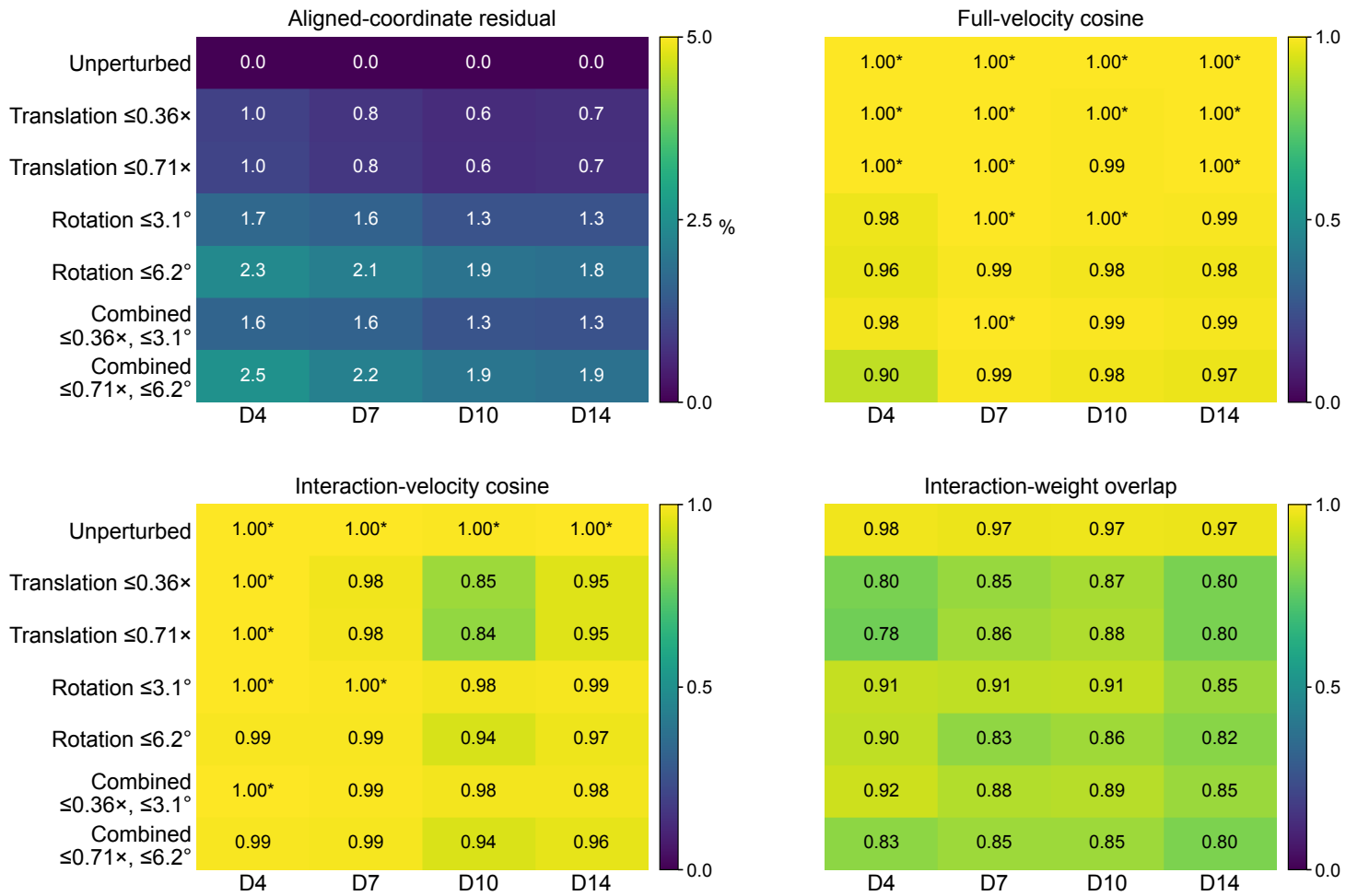
c Cell-state transitions



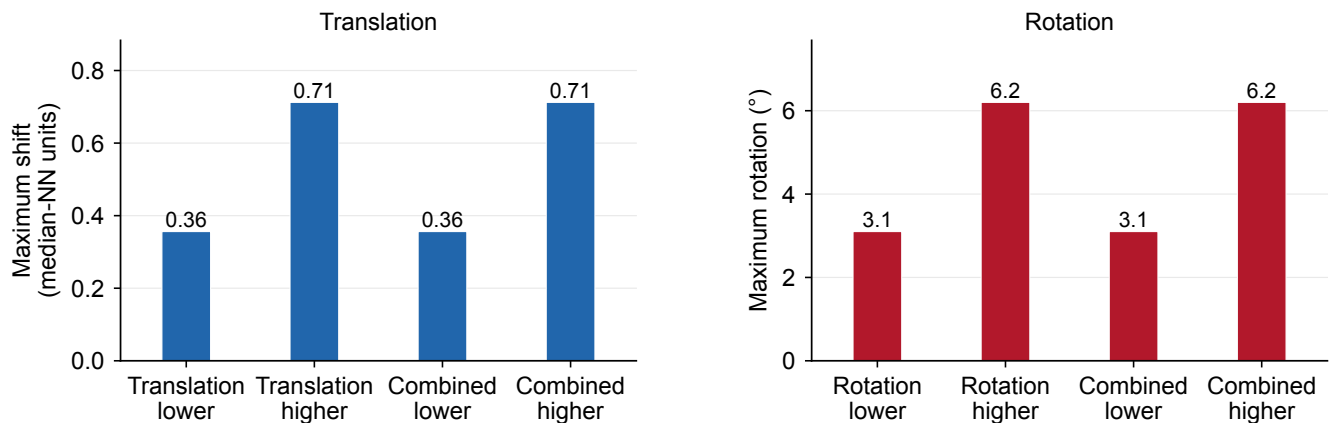
a Chicken-heart sections



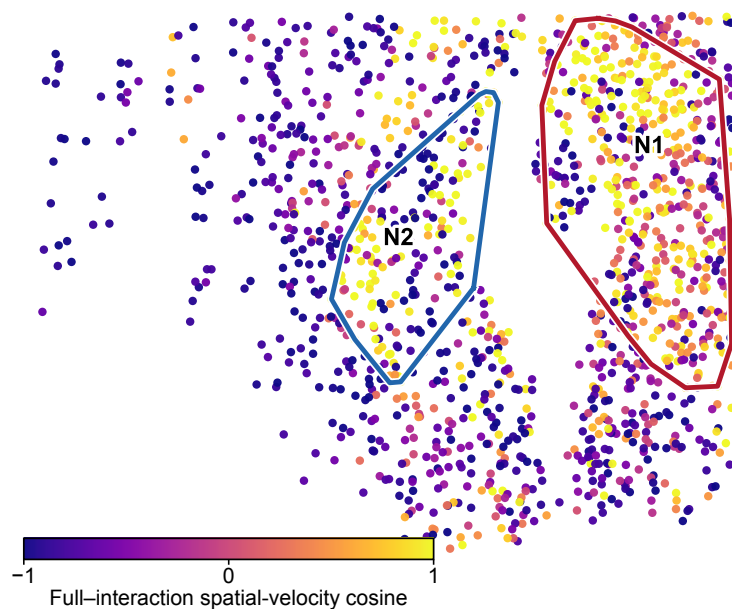
b Alignment and downstream sensitivity



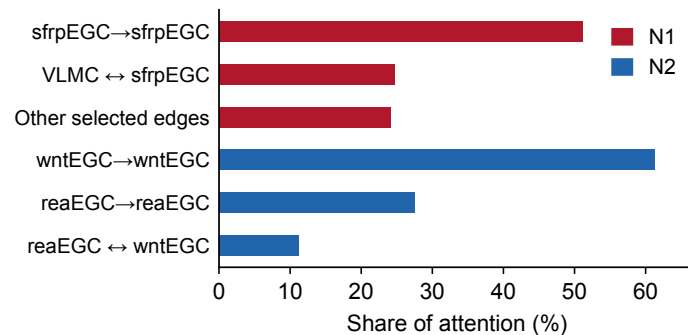
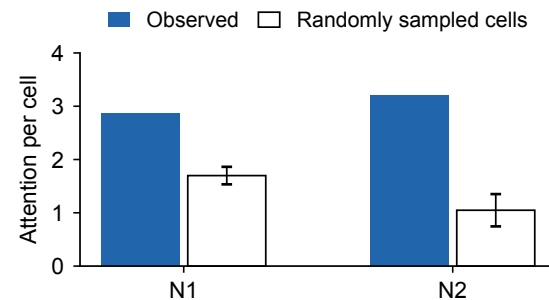
c Perturbation sizes



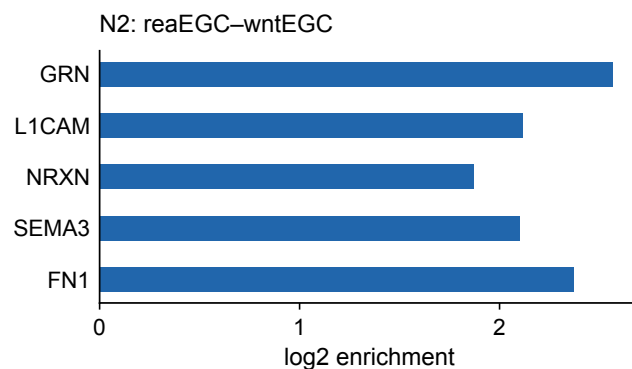
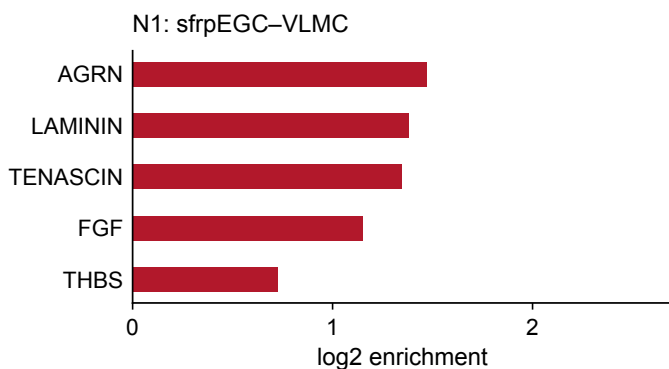
a Spatial interaction domains



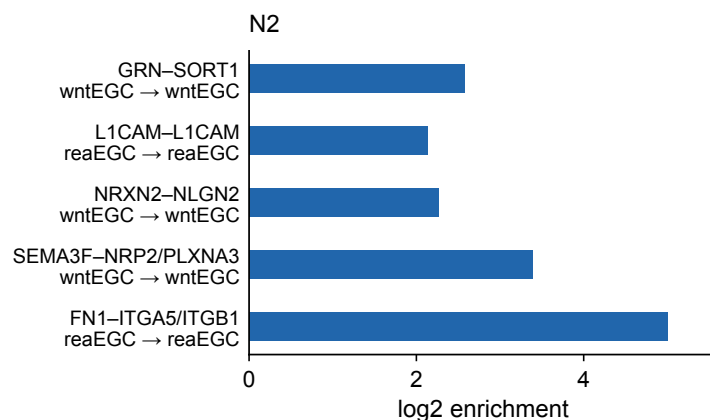
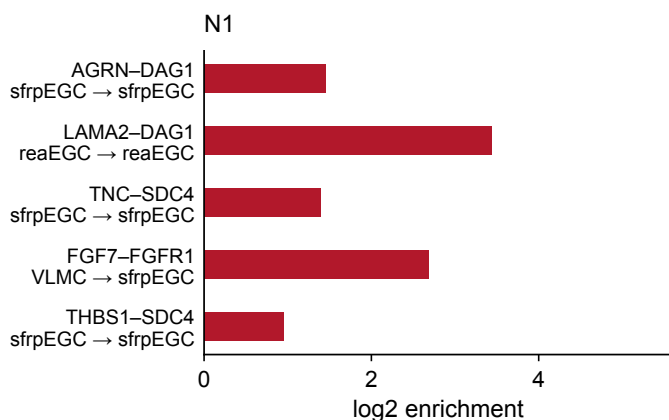
b Within-domain interactions



c Pathway enrichment



d Ligand–receptor pairs



a Spatial distributions

Baseline
n = 5,177



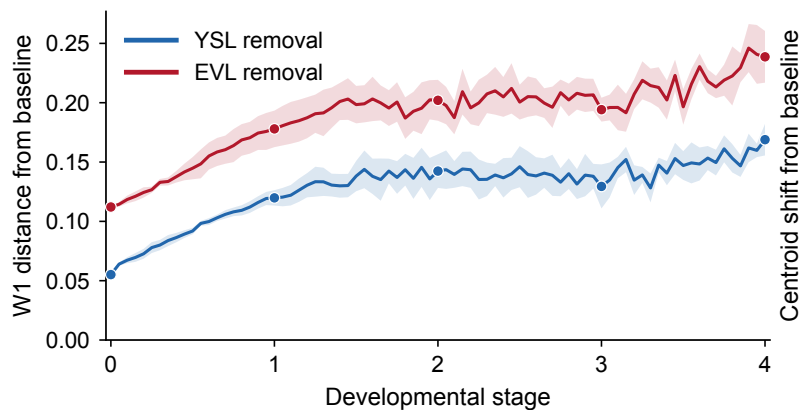
YSL removal
n = 4,546



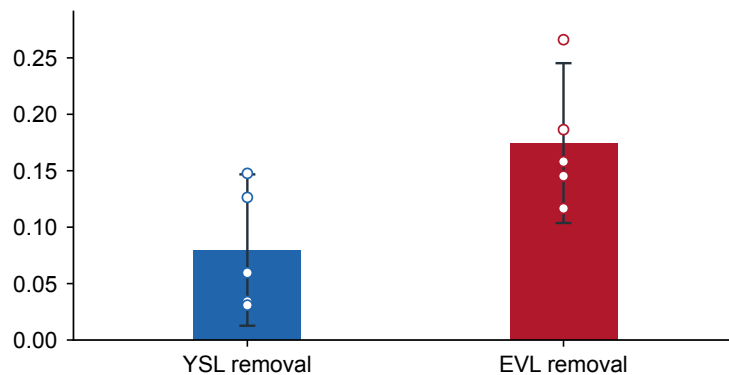
EVL removal
n = 3,306



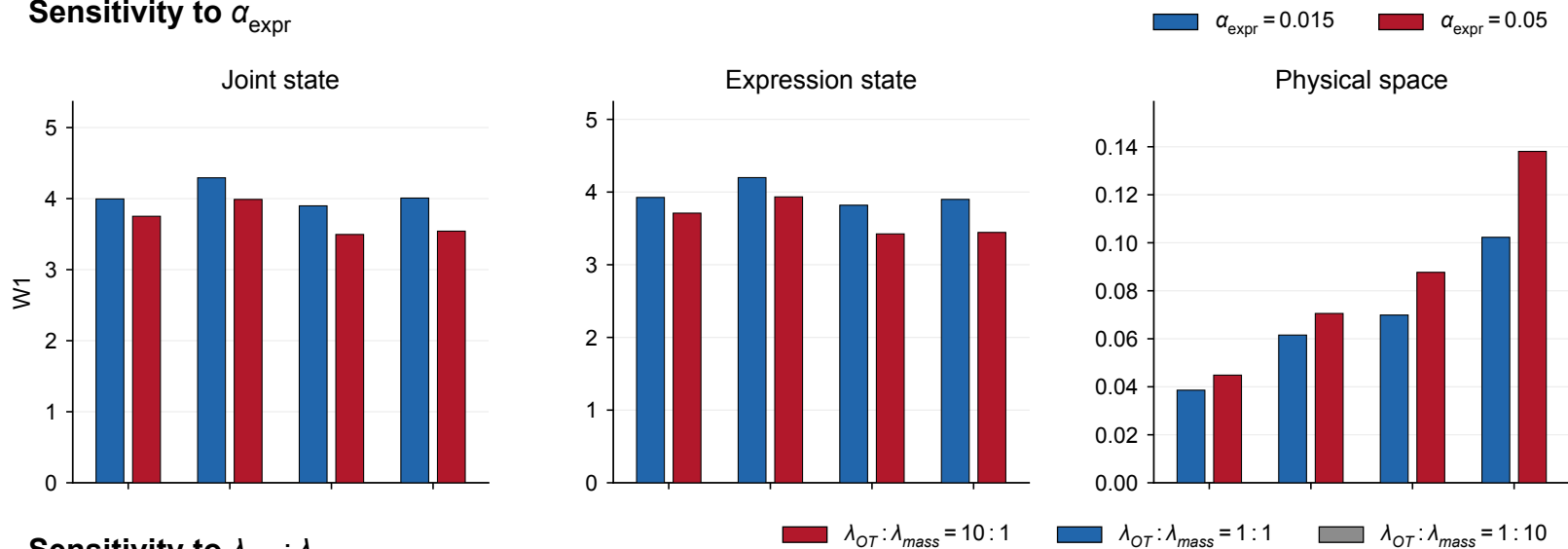
b Spatial W1



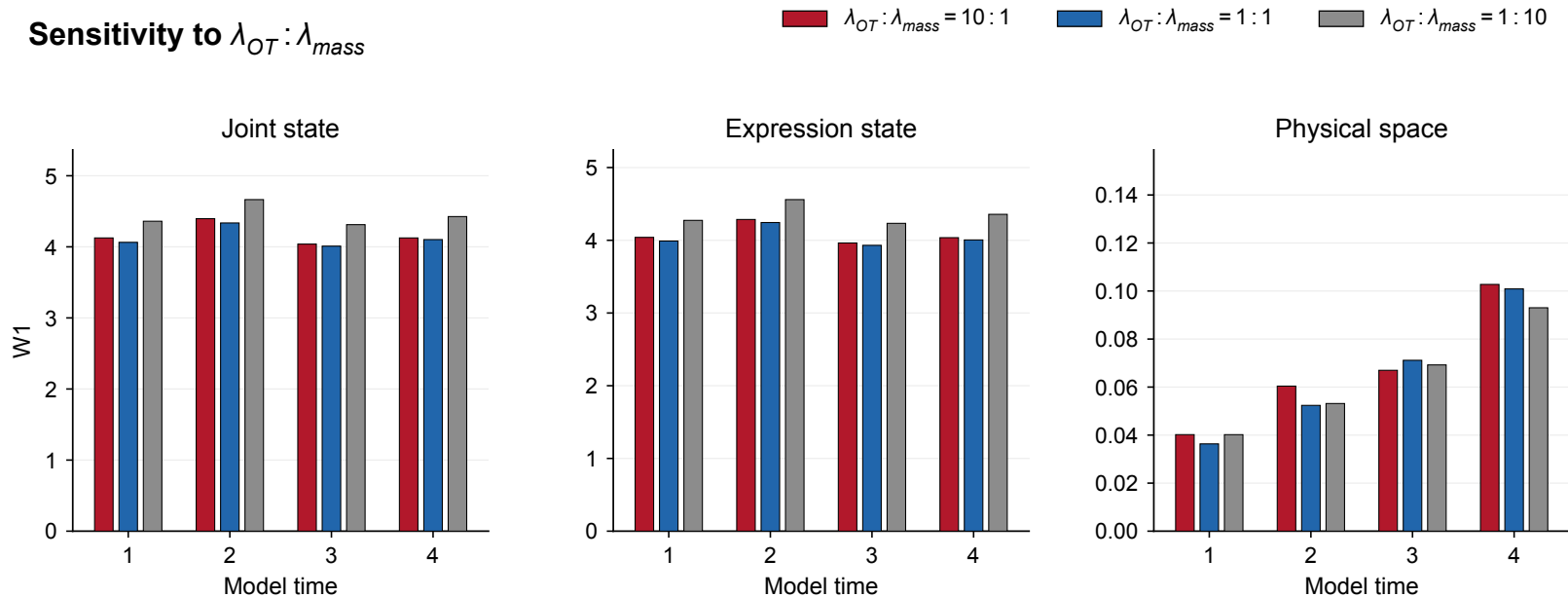
c Centroid shift



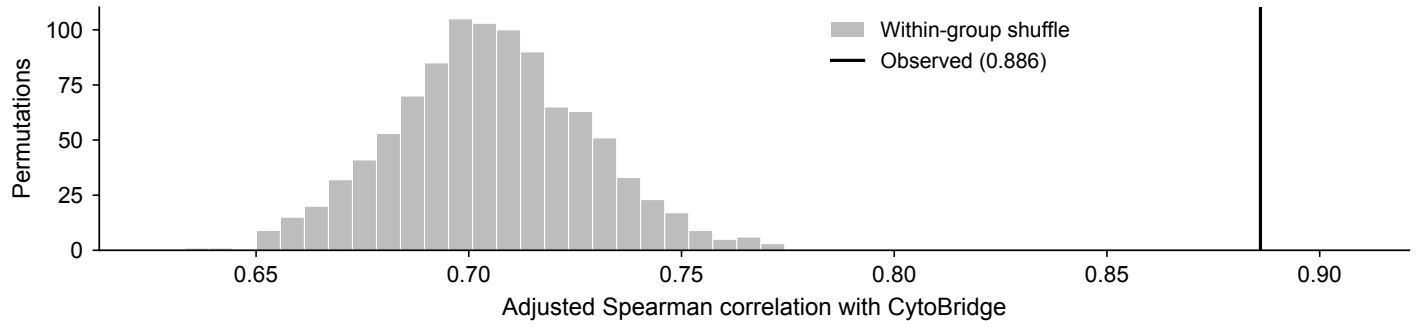
a Sensitivity to α_{expr}



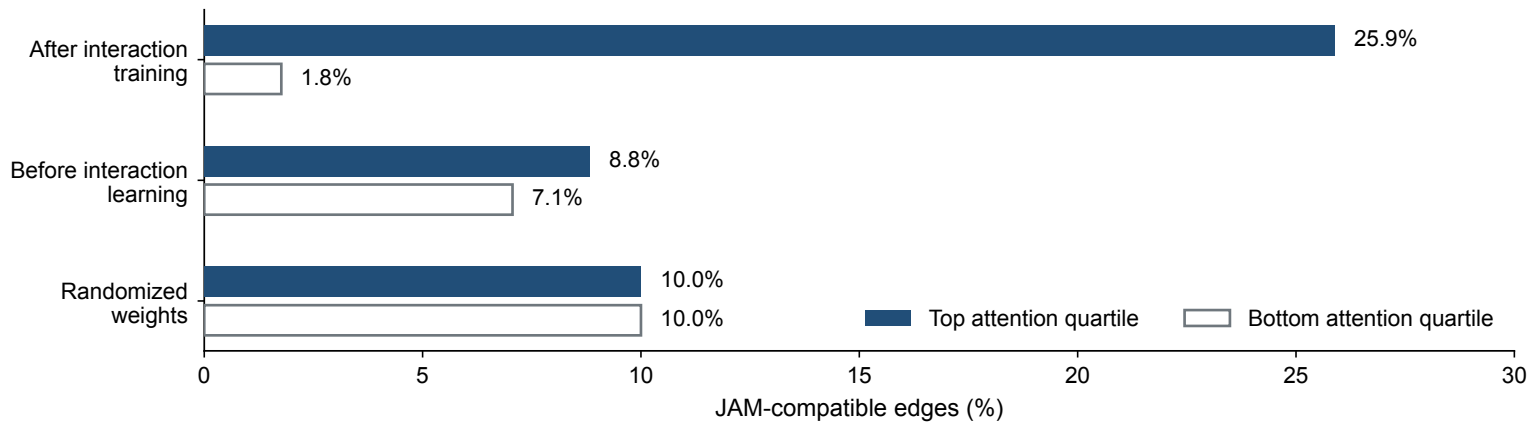
b Sensitivity to $\lambda_{OT}:\lambda_{mass}$



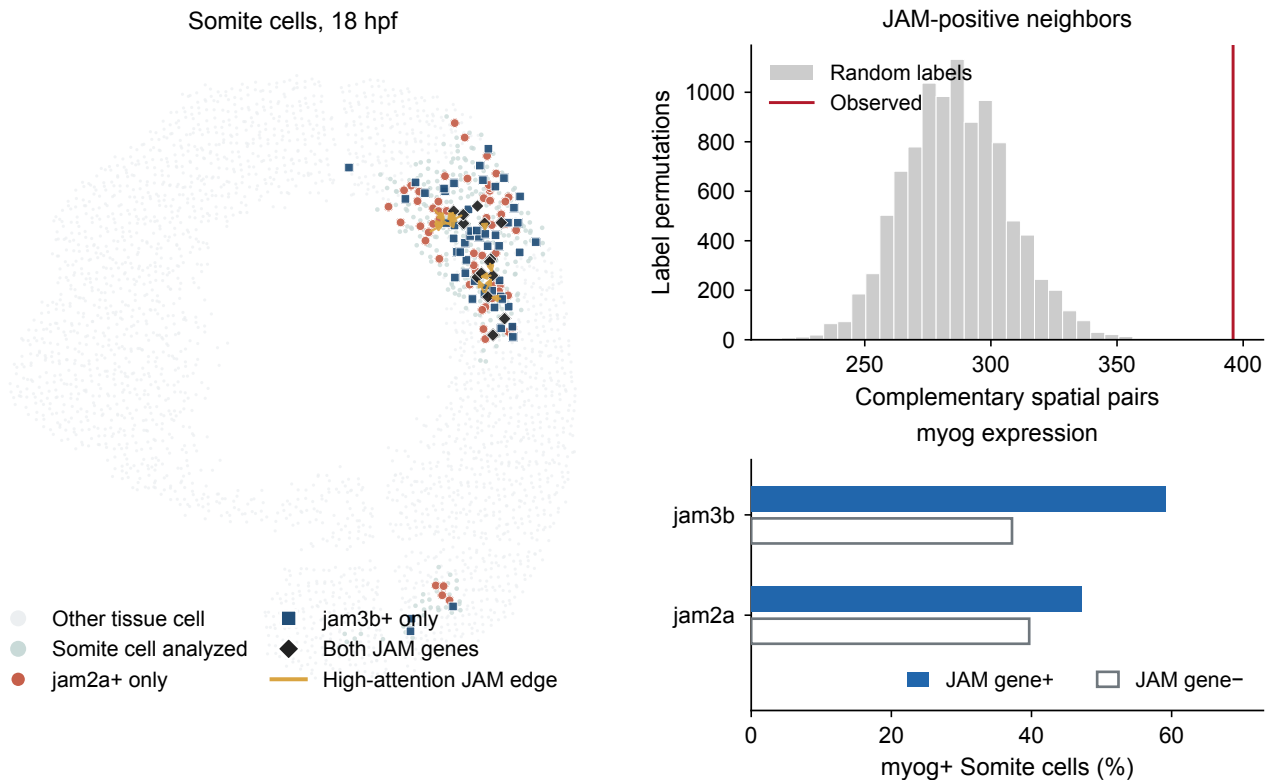
a COMMOT comparison



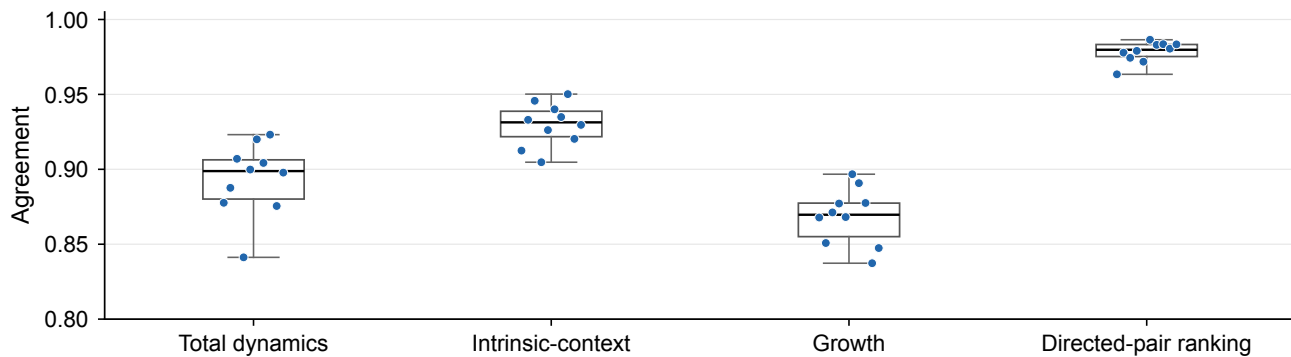
b JAM-compatible interactions



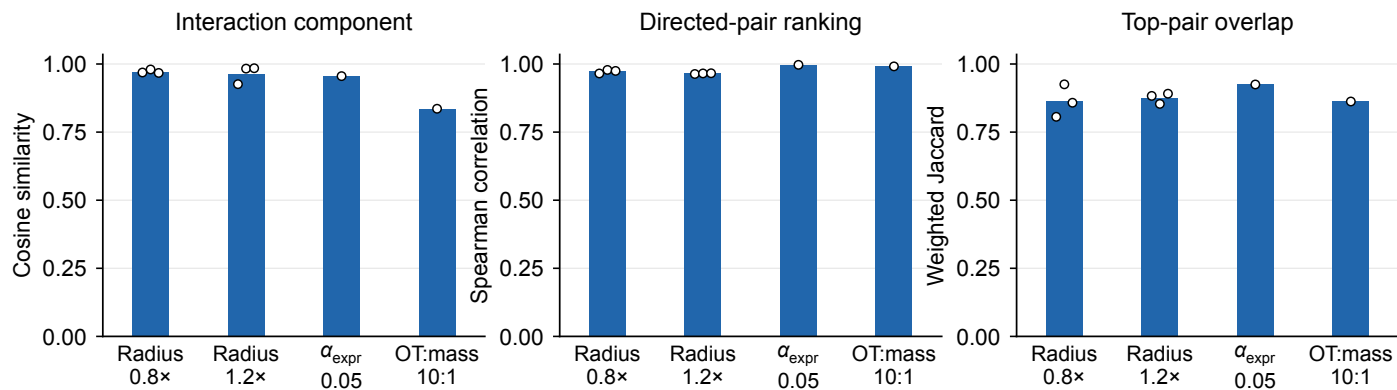
c JAM expression and spatial neighbors



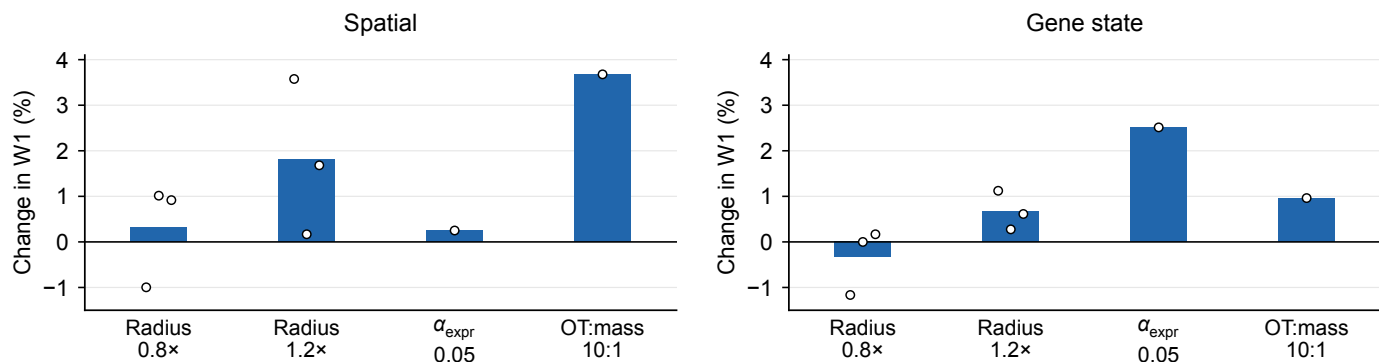
a Agreement across training seeds



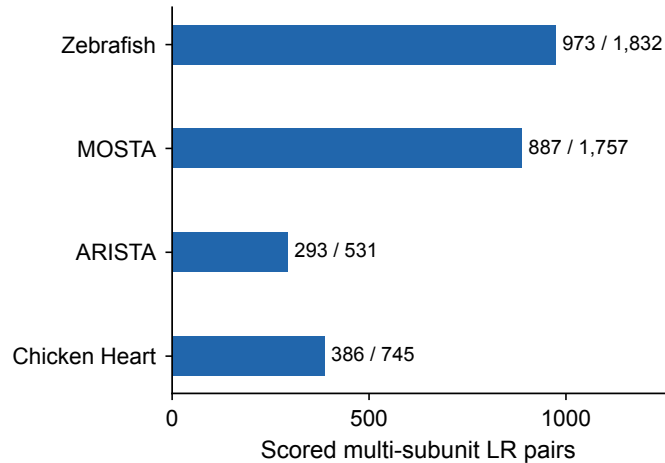
b Sensitivity to model settings



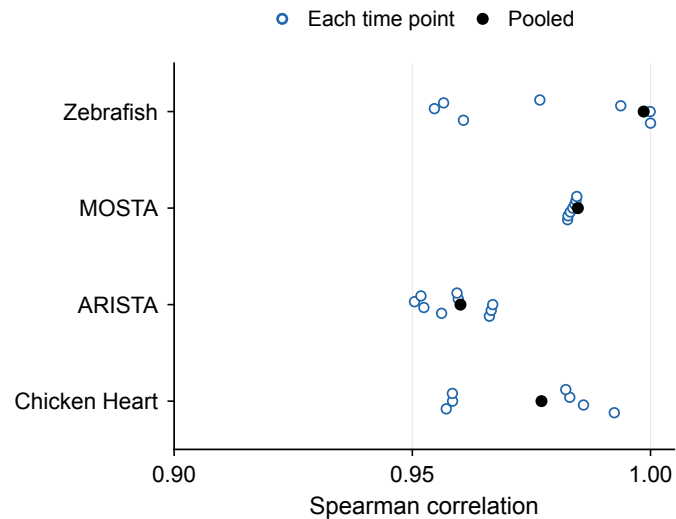
c Reconstruction error



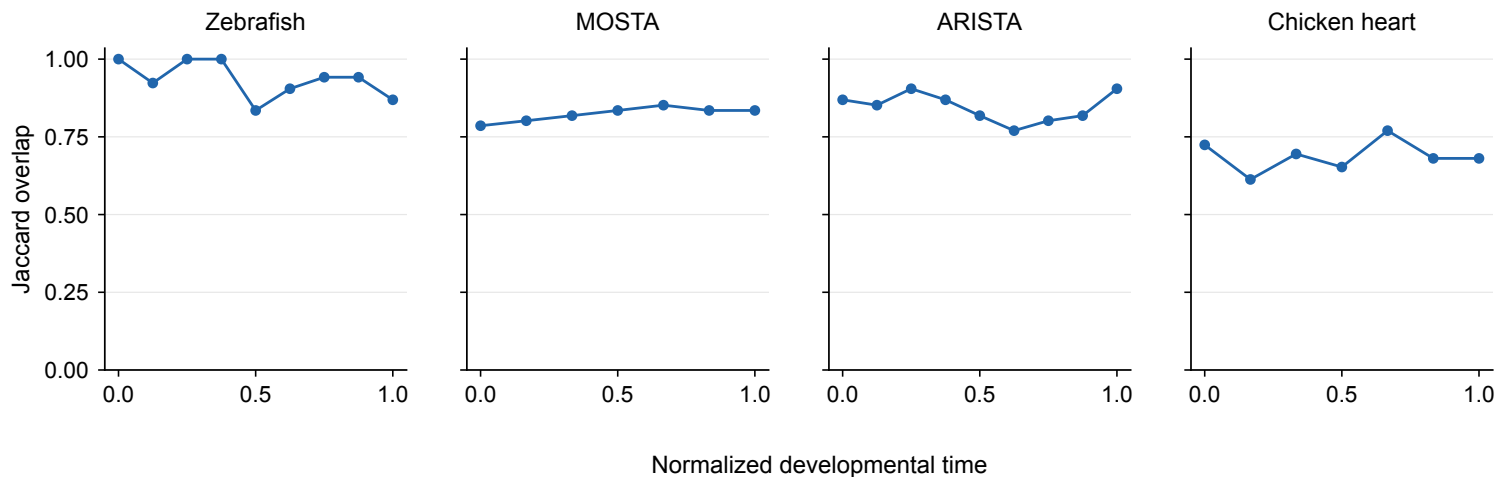
a Multi-subunit LR pairs



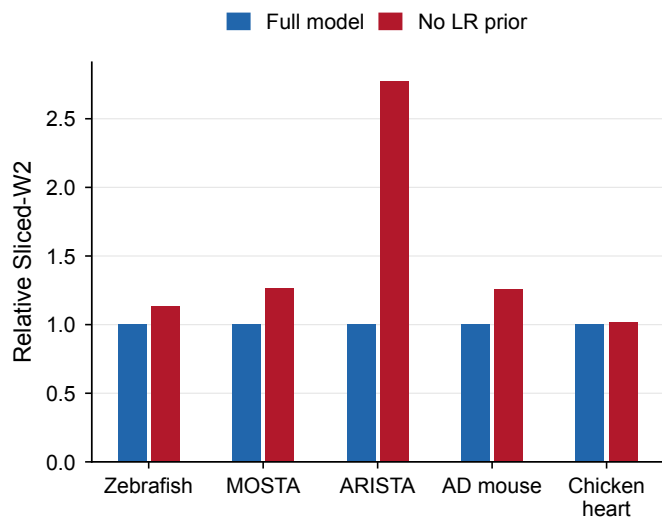
b LR-score agreement



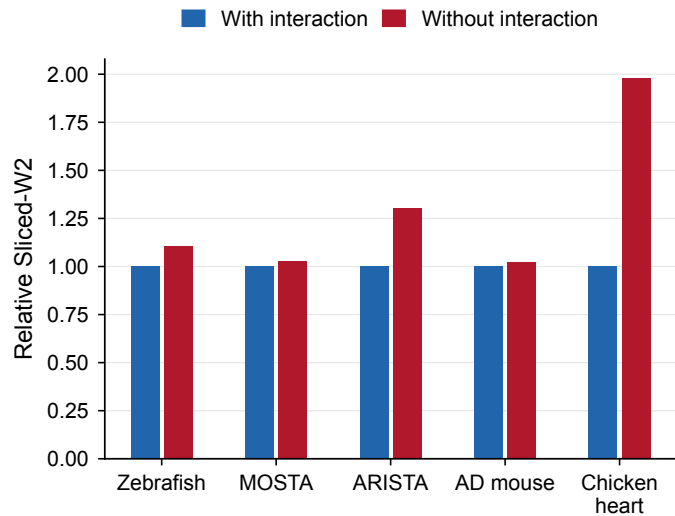
c Top-100 LR-pair overlap



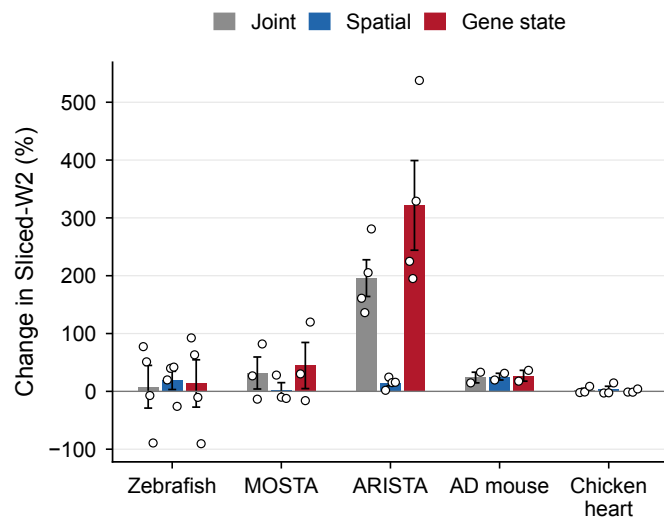
a LR-prior ablation



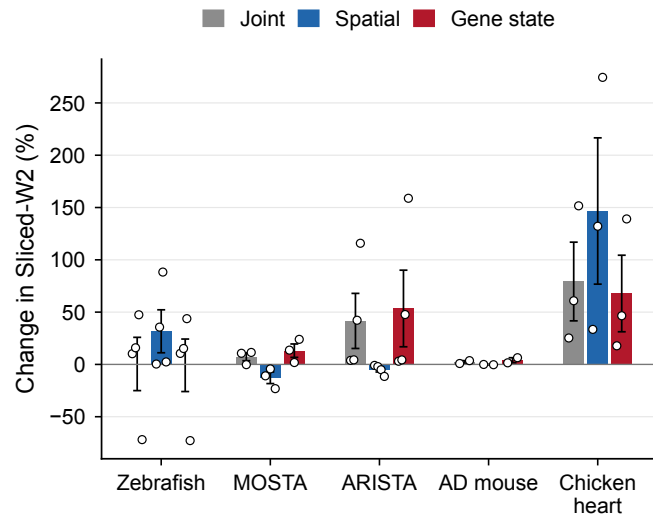
c Interaction ablation



b Reconstruction error

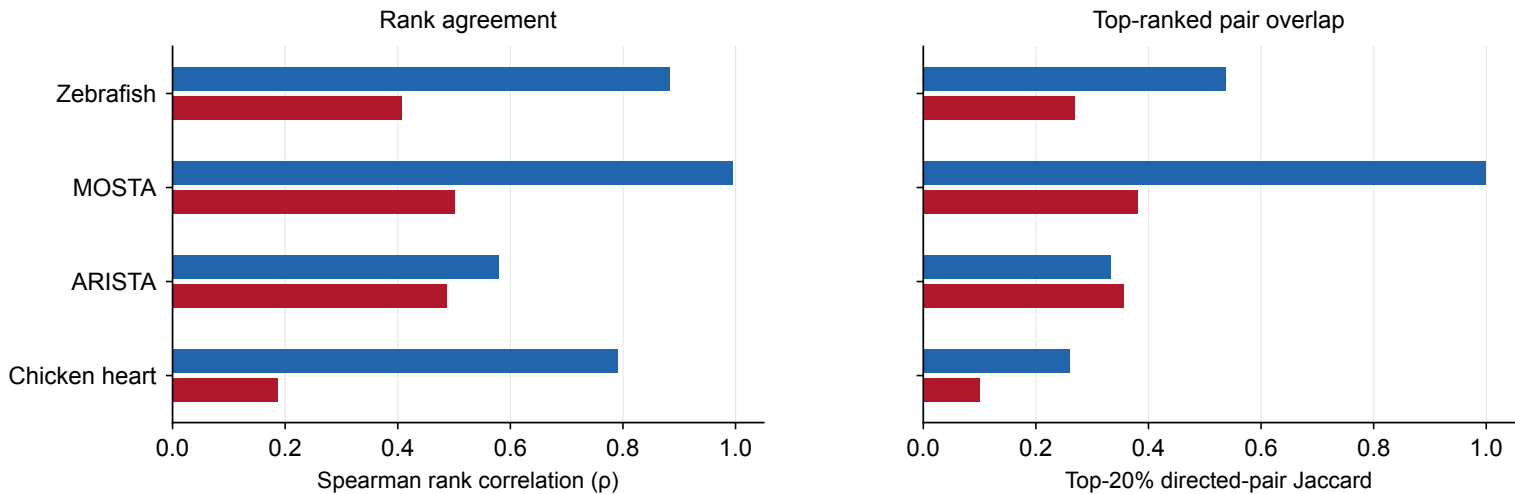


d Reconstruction error



a Directed-pair consistency

■ COMMOT ■ CellAgentChat



b CytoBridge-selected LR axes

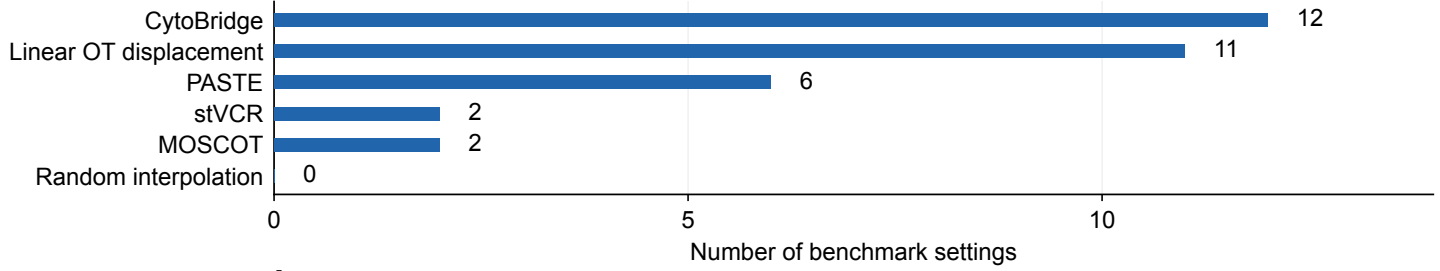
Dataset	CytoBridge selection				COMMOT comparison	
	Sender	Receiver	Selected LR axis	Database pathway	Same-pair LR rank	Pair percentile
Zebrafish	Pronephros	Musculature / YSL	COL1A2–SDC4	COLLAGEN	7 / 1,991	75.9
MOSTA	Muscle	Connective tissue	COL6A3–CD44	COLLAGEN	44 / 1,757	93.3
ARISTA	sfrpEGC	tinBL	PSAP–GPR37L1	PSAP	2 / 480	94.2
Chicken heart	Fibroblast cells	Valve cells	CD99–CD99	CD99	2 / 1,105	96.5

c NicheNet receiver targets for CytoBridge-selected axes

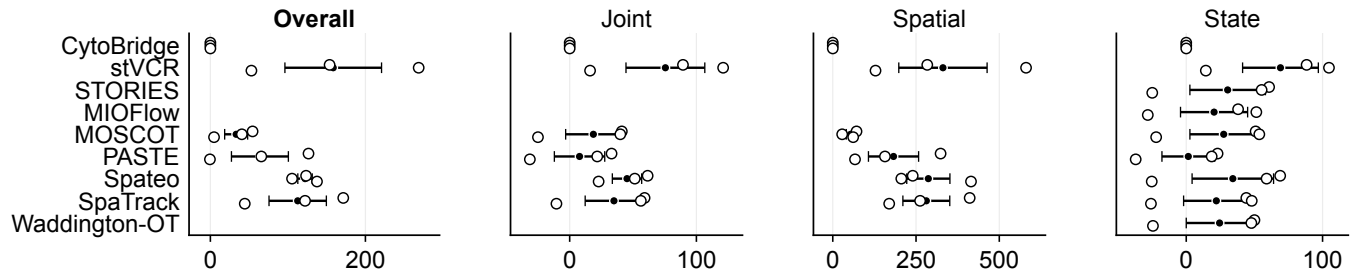
Dataset	CytoBridge selection		COMMOT and NicheNet results	
	Directed pair (sender; receiver)	LR axis Database pathway	COMMOT same-axis percentile	NicheNet-predicted receiver targets
Zebrafish	Sender: Spinal Cord Anterior Region Receiver: Nervous System	CDH2–CDH2 CDH	99.2	Ccnd1, Cbx5, Cdk1
MOSTA	Sender: Muscle Receiver: Connective tissue	COL6A3–CD44 COLLAGEN	99.3	Cd44
ARISTA	Sender: sstIN Receiver: mpEX	L1CAM–L1CAM L1CAM	99.1	EGR1, DDIT4, JUNB
Chicken heart	Sender: Valve cells Receiver: Fibroblast cells	LAMB1–DAG1 LAMININ	99.9	ID1, FN1, HNRNPR

○ Held-out time point —•— Mean $\pm$ s.e.m.

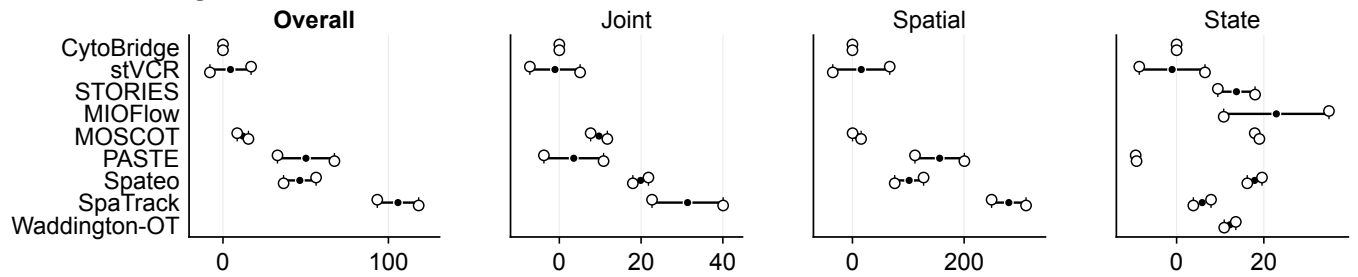
a Benchmark results across datasets



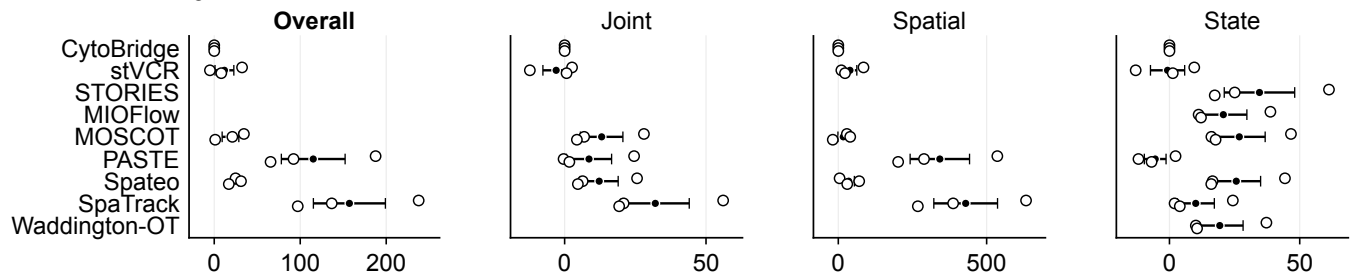
b Zebrafish



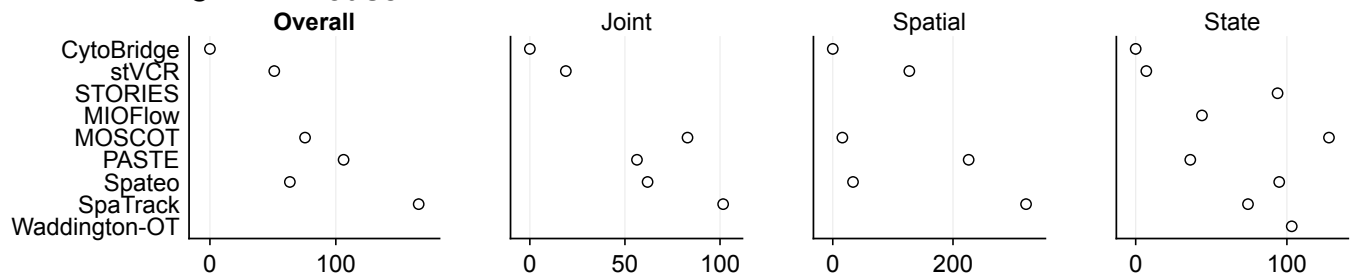
c MOSTA



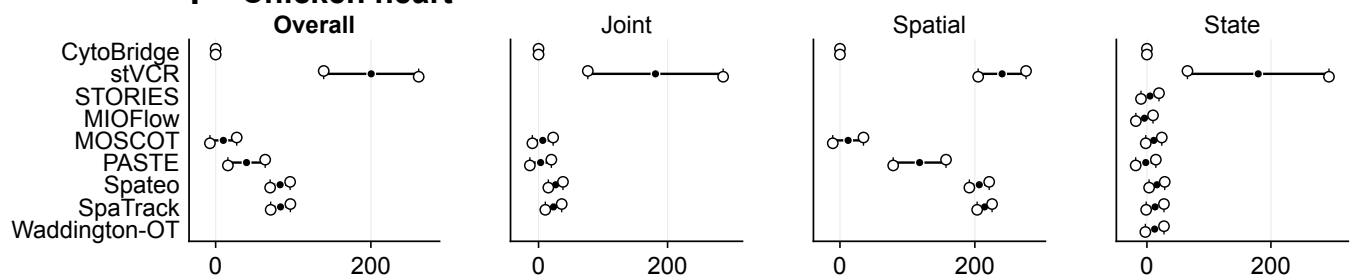
d ARISTA



e AD mouse

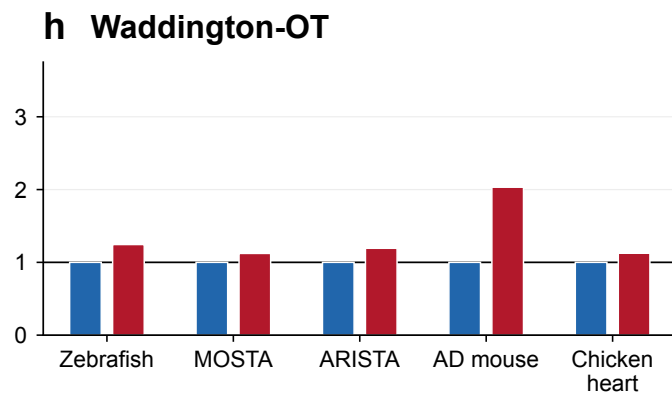
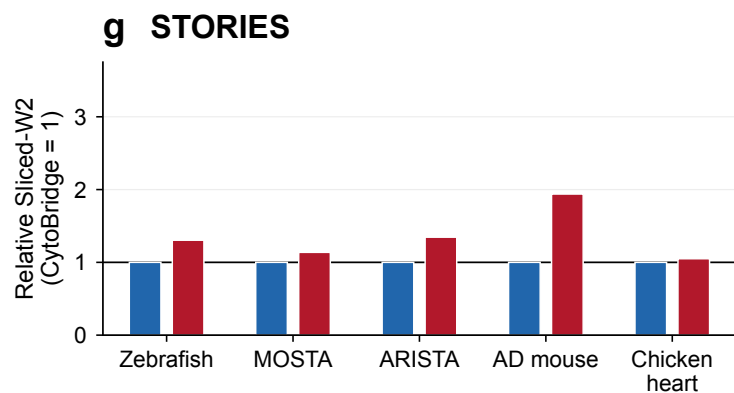
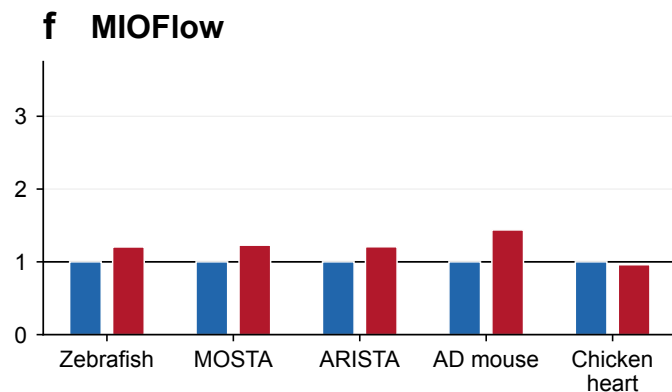
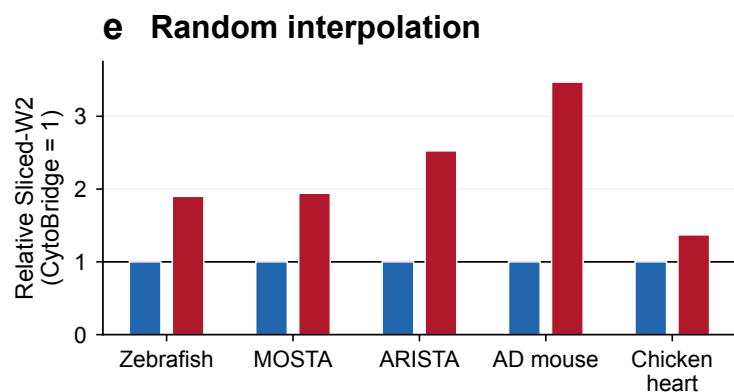
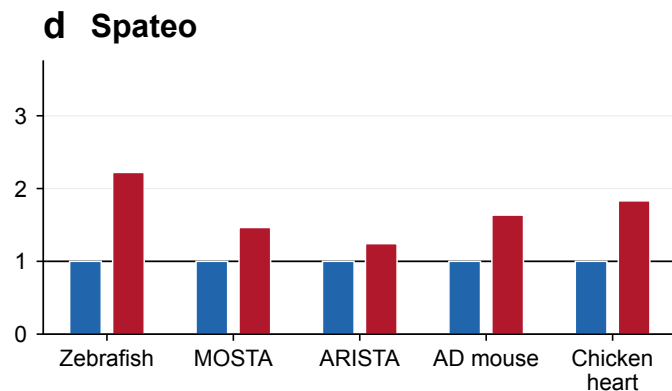
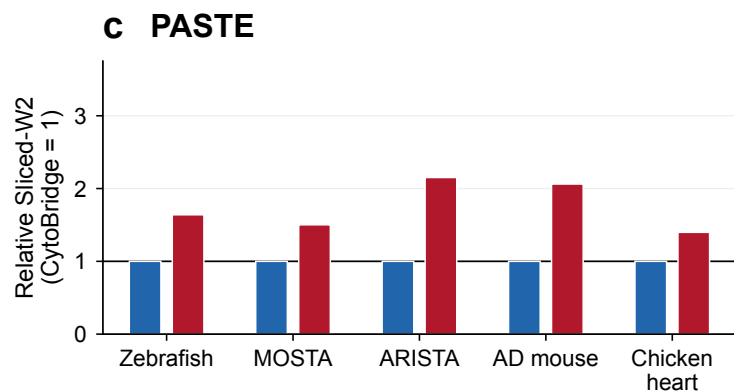
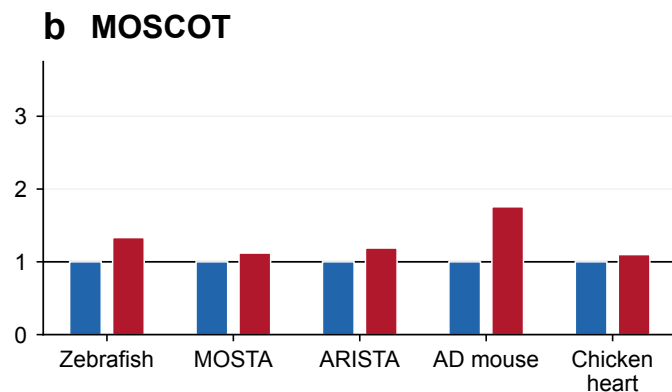
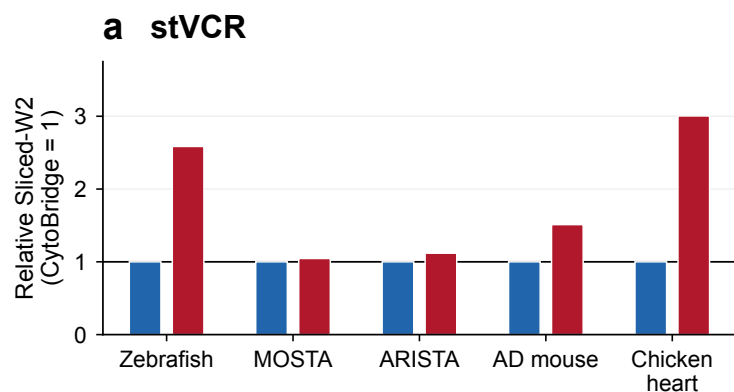


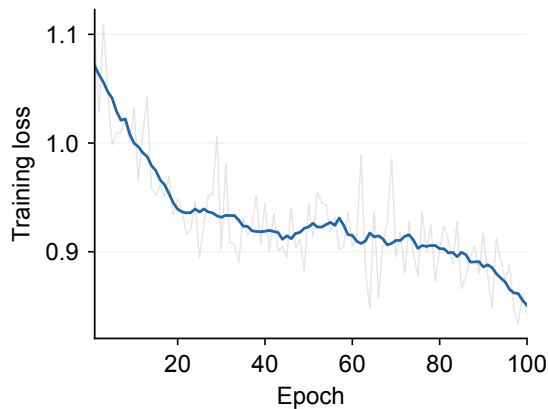
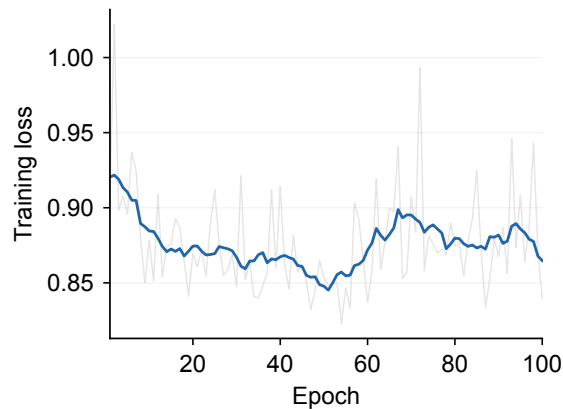
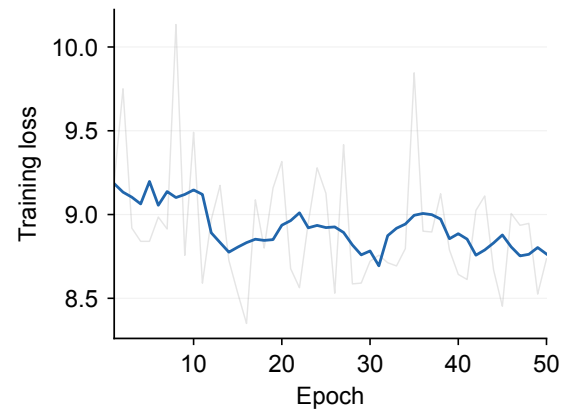
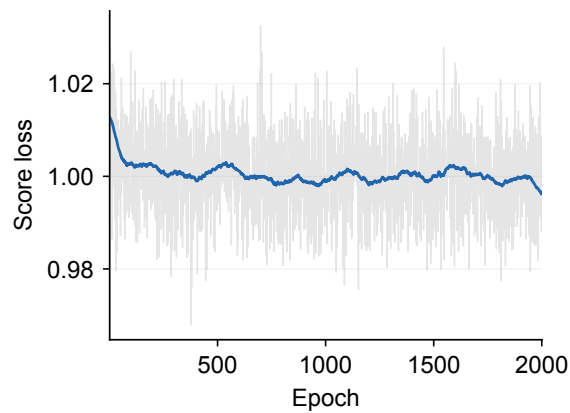
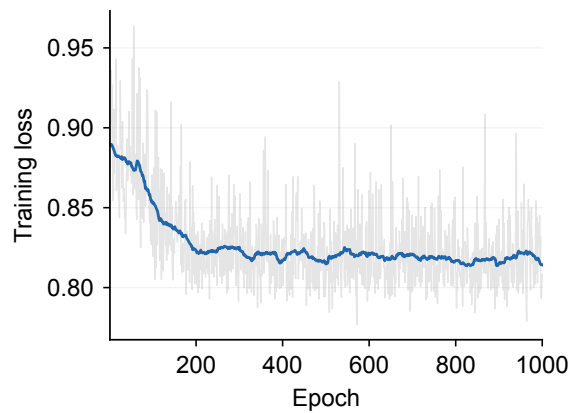
f Chicken heart



Relative error vs CytoBridge (%)

CytoBridge Comparison method



a Pretraining**b** Refinement**c** Interaction training**d** Score training**e** Fine-tuning**f** Score refinement